



國立陽明交通大學資訊工程學系資訊中心

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HW1-0 - Environment Setup

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Purpose

- The goal is to build two network zones
 - DMZ
 - VPN, Mail, WWW reverse proxy, etc.
 - Private zone
 - DHCP, DNS, LDAP, WWW backend etc.
- Know what you should know about the configuration and management of these services

Overview

- Router
 - The VM has direct Internet access
 - Provides NAT and Firewall
 - Able to connect all VMs in intranet
- Agent
 - Simulate simple VMs inside two zones separately help TA and OJ to verify results
- Client (Optional)
 - Any VMs you want for testing or running service

Definitions (1/2)

- OJ
 - Online Judge system, <https://nasaoj-v3.it.cs.nycu.edu.tw/>
- VPN server
 - A WireGuard server which connects your intranet so that OJ worker can access
- Internet
 - The IP addresses that are not in our intranet
- Intranet
 - A network zone including VPN zone, DMZ and your private zone which are managed by yourself

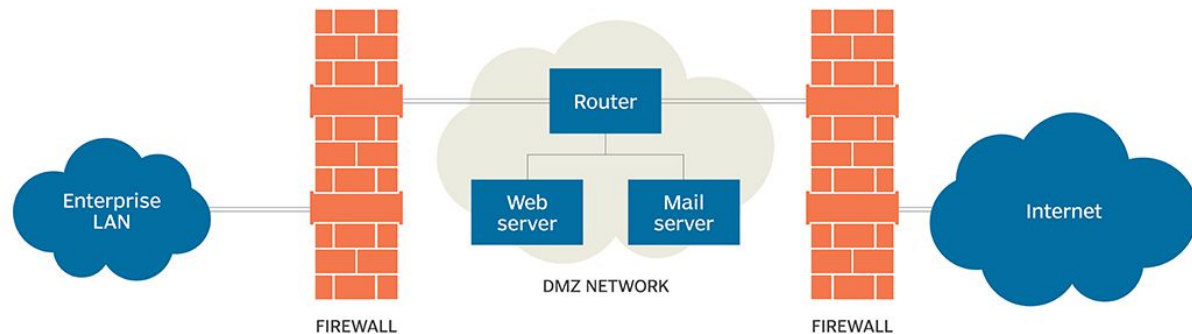
Definitions (2/2)

- VPN zone
 - A private network provided via WireGuard configuration for you and it can be accessed by online judge
- DMZ
 - 172.16.0.0/24, demilitarized zone
- Private zone
 - 172.16.1.0/24, a subnet of intranet managed by yourself
 - Shouldn't have direct access to Internet

What is DMZ?

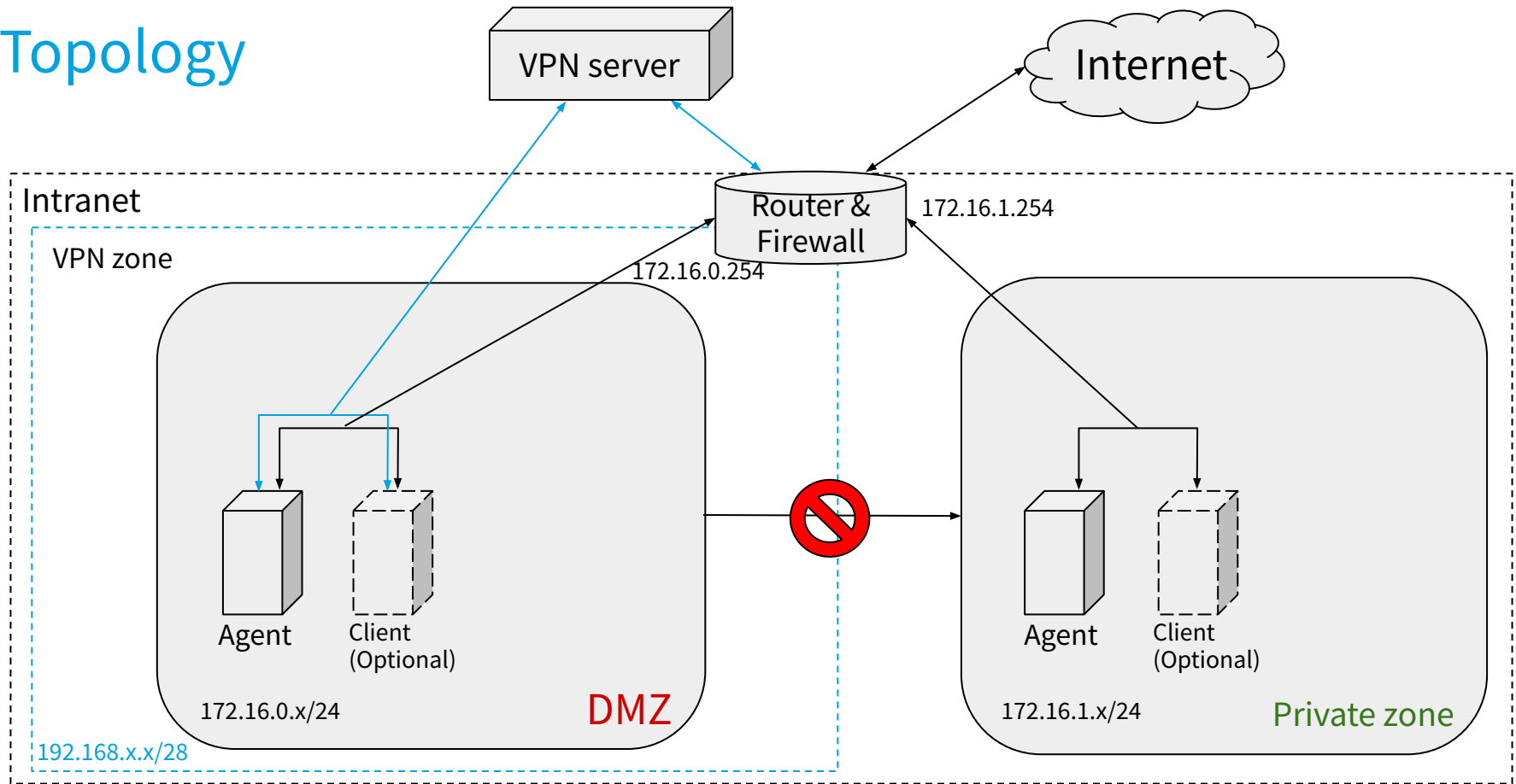
- The network zone between private zone and Internet
- Traffic from here to LAN should be monitored or checked by firewall

DMZ network architecture



Source: [What is a DMZ in Networking? | Definition from TechTarget](#)

Topology



Requirements

Routing (1/2)

- Router
 - Any OS is available
 - This VM must have these interfaces
 - External (For Internet access)
 - VPN zone (For OJ access)
 - **DMZ** (172.16.0.254/24)
 - **Private zone** (172.16.1.254/24)

Routing (2/2)

- All inbound and outbound traffic for your **Private zone** should go through Router
- Traffic from **Private zone** to Internet should be NAT masqueraded
- Set DNAT to both agents in **DMZ** and **Private zone**
 - Router's 10001 port should be mapped to **DMZ** agent's 2222 port
 - Router's 10002 port should be mapped to **Private zone** agent's 2222 port

Setup agent (1/2)

- Agent VM
 - OJ will login to **Agent containers** to judge your system
 - Setup **two VMs**
 - One is inside **DMZ** with IP 172.16.0.123
 - One is inside **Private zone** with IP 172.16.1.123
 - Install Docker for your Agent VMs
 - [Install | Docker Docs](#)

Setup agent (2/2)

- Agent is a Docker container inside your VM
 - Container image download link:
<https://nextcloud.it.cs.nycu.edu.tw/s/JGbHmEe5Pz5PxDm>
 - Start the Docker container with specified image
 - This will enable Agents' 2222 and 55688 ports for connection
 - Please **don't** revise any settings inside the container!

```
$ wget https://nextcloud.it.cs.nycu.edu.tw/public.php/dav/files/JGbHmEe5Pz5PxDm -O agent.tar
$ docker load < agent.tar
$ docker run -d --restart unless-stopped --name nap-agent --network host nap-agent
```

Firewall

- Configure firewall rules on Router
- Rules
 - By default, all connections from other zones to **Private zone** should be rejected
 - By default, all connections from **Private zone** to VPN zone should be rejected
 - **DMZ** to **Private zone** should be allowed for 55688 port
 - ICMP connections from anywhere to anywhere are allowed
 - Internet connection of **DMZ** and **Private zone** should be allowed
 - SSH connections from VPN zone to Router should be rejected
 - SSH connections from Router to both Agents should be allowed

OJ checkpoints

1. Router's VPN connection
2. Router's DNAT to Agents in two zones
3. Two Agents' Internet connection
4. Trace route from Agent in **Private zone**
5. Check firewall rules for Router
6. Check firewall rules for **Private zone**

Help me!

- Previous SA slides
 - [SA - 2025 課程內容 | NASA Course Website](#)
- Appendix in previous NA slides
 - <https://site.nasa.cs.nycu.edu.tw/na/2024/HW1.pdf>
- How To Ask Questions The Smart Way
 - <https://github.com/ryanhanwu/How-To-Ask-Questions-The-Smart-Way>

Good Luck!